

Decode lab

Project #02:

Automated Irrigation Controller (Actuator Logic)

Introduction:

The purpose of this project is to build an automated irrigation system using IoT concepts. The system reads soil moisture values using a Soil Moisture Sensor and automatically controls a water pump through a relay module.

This project demonstrates:

- Analog-to-Digital Conversion (ADC)
- Threshold-based decision making
- Relay actuator control
- Closed-loop automation

Objective:

To design an automated irrigation controller that:

- 1.Reads analog moisture values

2.Detects dry soil conditions

3.Activates relay and water pump automatically

4.Stops the pump when soil becomes wet

Required Components:

Component

Quantity

UNO

1

Soil Moisture Sensor

1

5V Relay Module

1

Water Pump / LED

1

Breadboard

1

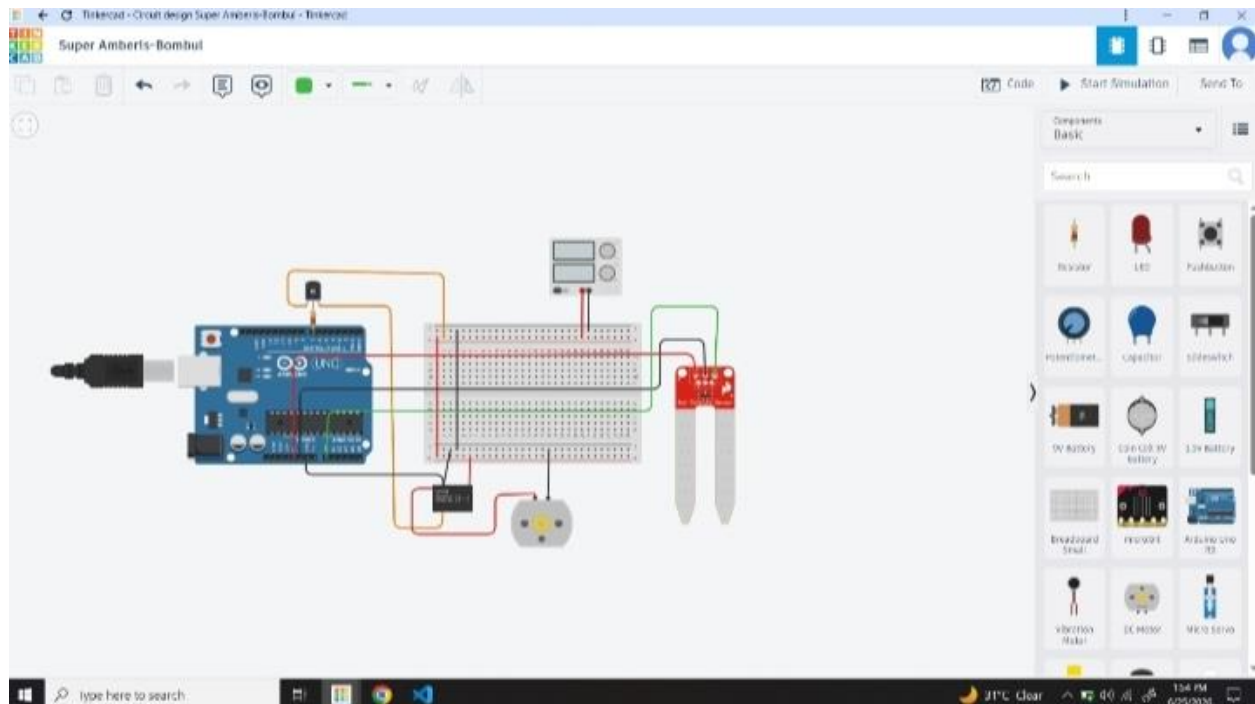
Jumper Wires

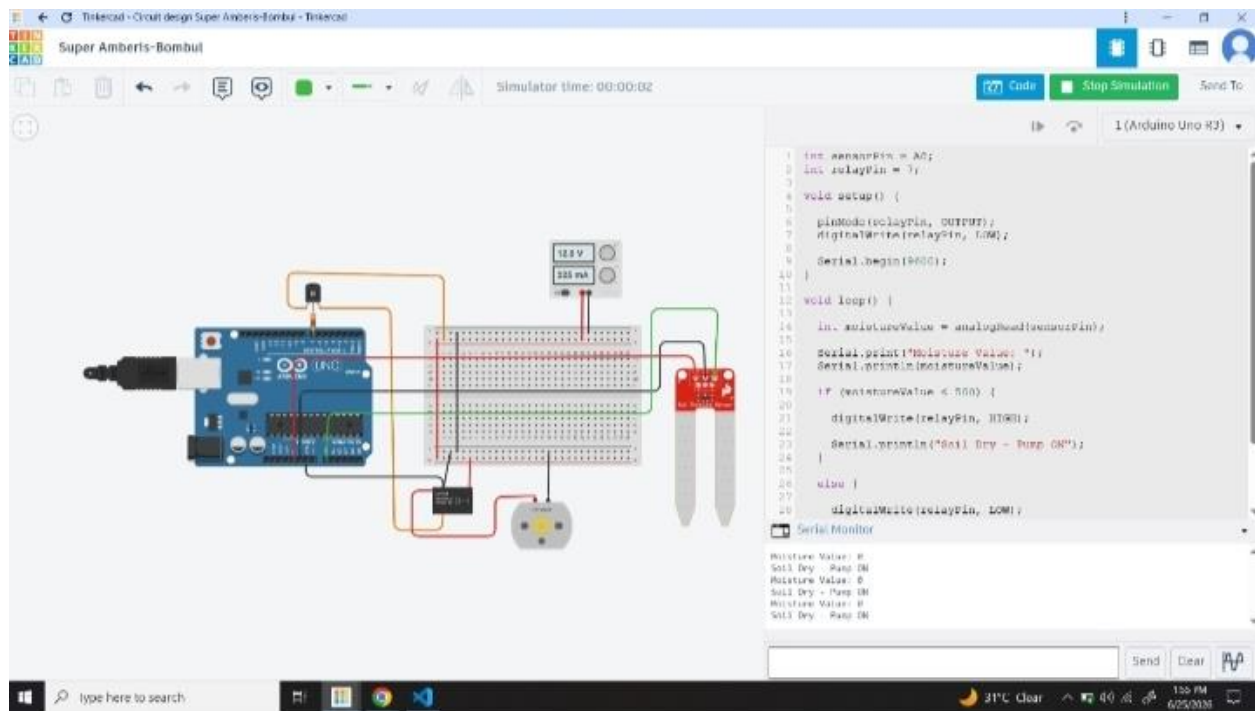
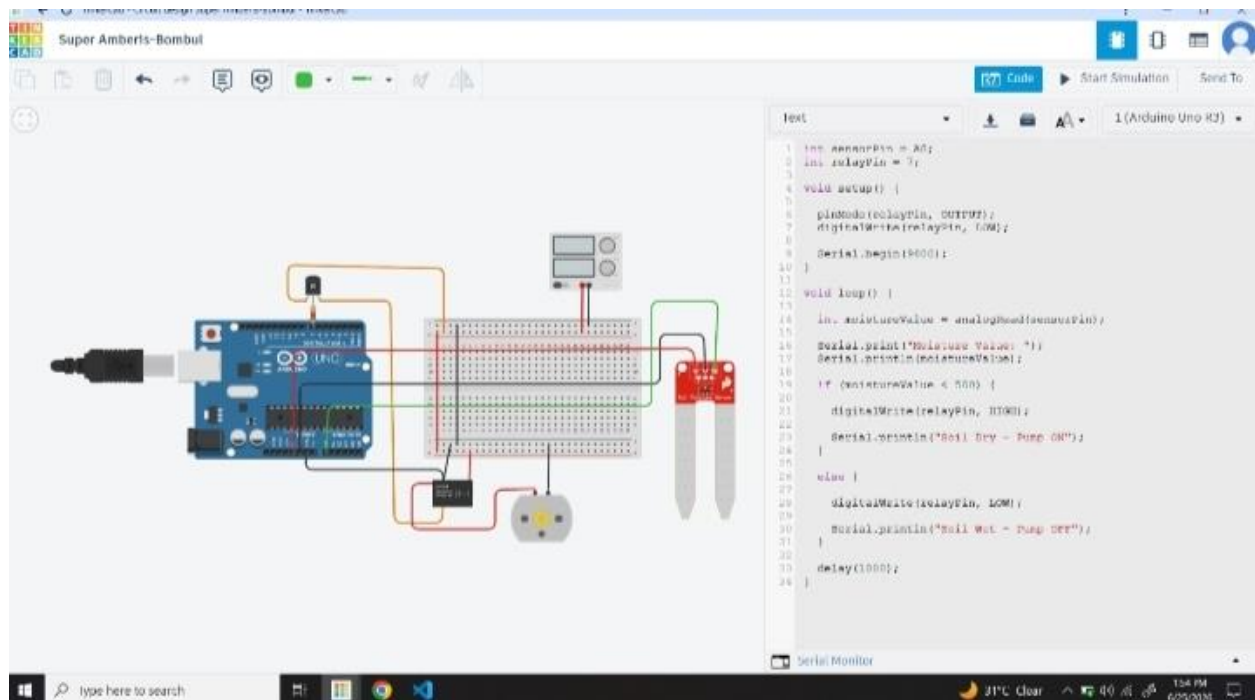
Few

USB Cable

1

Circuit Diagram:





Circuit Connections:

Soil Moisture Sensor Connections

Sensor Pin

Arduino Pin

VCC

5V

GND

GND

A0

A0

Relay Module Connections:

Relay Pin

Arduino Pin

VCC

5V

GND

GND

IN

D7

Pump Connections:

- Pump positive wire → Relay NO
- Relay COM → External power supply positive
- Pump negative wire → Power supply negative

Arduino Code:

```
int sensorPin = A0;
```

```
int relayPin = 7;
```

```
void setup() {
```

```
    pinMode(relayPin, OUTPUT);
```

```
    digitalWrite(relayPin, LOW);
```

```
    Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
int moistureValue = analogRead(sensorPin);
```

```
Serial.print("Moisture Value: ");
```

```
Serial.println(moistureValue);
```

```
if (moistureValue < 500) {
```

```
    digitalWrite(relayPin, HIGH);
```

```
    Serial.println("Soil Dry - Pump ON");
```

```
}
```

```
else {
```

```
    digitalWrite(relayPin, LOW);
```

```
    Serial.println("Soil Wet - Pump OFF");
```

```
}
```

```
delay(1000);
```

```
}
```

Working principal:

1. Soil moisture sensor continuously reads soil condition.
2. Arduino converts analog signals into digital values using ADC.
3. Threshold logic compares moisture value.
4. If soil is dry:
 - ° Relay activates
 - ° Water pump turns ON
5. If soil is wet:
 - ° Relay deactivates
 - ° Water pump turns OFF

Flowchart:

Start

↓

Read Soil Moisture Sensor

↓

Is Moisture < Threshold?

↓ Yes

↓ No

Relay ON

Relay OFF

Pump ON

Pump OFF

↓

↓

Repeat Process Continuously

Output Results:

Dry Soil Condition

Moisture Value: 320

Soil Dry – Pump ON

Wet Soil Condition:

Moisture Value: 720

Soil Wet – Pump OFF

Key Concepts Used:

Concept

Description

ADC

Converts analog signals into digital values

Threshold Logic

Checks dryness condition

Relay Module

Controls high-power devices

Closed-loop Automation

Automatic response based on sensor input

Advantages:

- Saves water
- Fully automatic
- Reduces manual work
- Useful for smart agriculture
- Low-cost irrigation solution

Applications:

- Smart Farming
- Home Gardening

- Greenhouse Monitoring
- Agricultural Automation

Conclusion:

The Automated Irrigation Controller successfully automates watering operations using sensor-based decision making. The system continuously monitors soil moisture and controls the pump automatically through relay switching. This project demonstrates practical implementation of IoT actuator control and embedded automation systems.